KESTERLINE ENGINEERING, PC
Consulting Structural and Geotechnical Engineers
215 South Third Street, Suite 402, Columbus, Ohio 43215 | (614) 555-0231 | kesterlineeng.com

ENGINEERING SUBMITTAL SUPPLEMENT

Date:	07/13/2026 (July 13, 2026)	**Letter Reference:**	KE-L-2026-0713
To:	Mr. Adrian Vance, Project Engineer (Structural/Geotechnical) Thelbridge Project Management Services, Owner's Representative	**Project:**	Birchwood Village Expansion Program, Package A
cc:	Mr. Steven Kendrow, Project Executive, Rowan Bay Constructors, Inc. Mr. Gregory Nash, PE, Calloway Braddock Structural Engineers, LLC	**Package / Facility:**	Package A / Block B Substructure
Subject:	**Submittal 31-SH-B Rev.1 — Revision 2 Engineering Supplement, Block B Plant-Room Support of Excavation**		

1. PURPOSE, RECORD REFERENCES & SUPERSESSION SCOPE

Kesterline Engineering, PC submits this Revision 2 Engineering Supplement (Letter Reference KE-L-2026-0713, dated 07/13/2026) to supplement **Submittal 31-SH-B Rev.1** for the Block B plant-room support of excavation (SOE). This supplement has been prepared in accordance with the project record documents, specifically:

- Geotechnical Investigation Report **KGI-2024-0417 Rev.2** (the GIR);
- Design Basis Memorandum **CB-DBM-001 Rev.2** (the DBM);
- Specification **Section 31 20 00 Rev.3** (Earthwork and Support of Excavation);
- **Submittal 31-SH-B Rev.1** (Block B Plant-Room Support of Excavation Design Package); and
- The **Package A register**.

The Contractor has confirmed that its Dewatering and Groundwater Control Plan, **Submittal 31-DW-A Rev.1**, is limited to the Block A basement and does not extend to the Block B plant-room excavation. Rev.1 of these calculations assumed a dry excavation on the basis of that plan. This Revision 2 supplement withdraws that assumption and establishes updated design calculations for active aquifer depressurization, excavation base heave, layered effective-stress plus hydrostatic lateral loading, sheet-pile toe embedment, tieback ground anchors, sheet-pile flexural adequacy, and wall deflection instrumentation triggers.

Supersession Statement: This Revision 2 supplement supersedes Submittal 31-SH-B Rev.1 Sections **3.4, 4.3, 5, 6, 7, 9, 10**. Submittal 31-SH-B Rev.1 Sections **1, 2, 3.1 to 3.3, 4.1, 4.2, 8, 11.3, 11.4, 12, 13** remain in force without modification.

Table 1 — Submittal 31-SH-B Rev.1 Section Supersession Matrix

Rev.1 Section	Rev.1 Section Title	Rev.2 Status	Governing Provision
Sections 1, 2	Project Scope, Codes & Site Reference Documents	Remains in Force	Submittal 31-SH-B Rev.1
Sections 3.1 to 3.3	Subsurface Stratigraphy & Soil Design Parameters	Remains in Force	Submittal 31-SH-B Rev.1
Section 3.4	Groundwater & Dry-Excavation Assumption	**Superseded**	Supplement Section **R2.1**
Sections 4.1, 4.2	Surface Surcharge & Construction Equipment Loads	Remains in Force	Submittal 31-SH-B Rev.1
Section 4.3	Lateral Earth Pressure Diagram Above Subgrade	**Superseded**	Supplement Section **R2.3**
Section 5	Excavation Base Stability Verification	**Superseded**	Supplement Section **R2.2**
Section 6	Sheet-Pile Toe Embedment Analysis	**Superseded**	Supplement Section **R2.4**
Section 7	Tieback Ground Anchor Design	**Superseded**	Supplement Section **R2.5**
Section 8	Global Overturning & Sliding Stability Checks	Remains in Force	Submittal 31-SH-B Rev.1
Section 9	Sheet-Pile Structural Bending & Shear Design	**Superseded**	Supplement Section **R2.6**
Section 10	Wall Deflection & Instrumentation Triggers	**Superseded**	Supplement Sections **R2.1 & R2.7**
Sections 11.3, 11.4	Waler, Bearing Plate & Anchor Head Details	Remains in Force	Submittal 31-SH-B Rev.1
Sections 12, 13	Construction Sequence Notes & Submittal Drawings	Remains in Force	Submittal 31-SH-B Rev.1

2. REVISION 2 ENGINEERING CALCULATIONS & SPECIFICATIONS (SECTIONS R2.1 – R2.8)

R2.1 — Groundwater Conditions, Relief Wells & Excavation Staging *(Supersedes Rev.1 Section 3.4 & Part of Section 10)*

The Rev.1 dry-excavation assumption is hereby formally withdrawn. For the Block B plant-room excavation (ground surface at El. 750.0, final excavation subgrade at El. 726.0, exposed wall height H = 24.0 ft), the revised design groundwater and stratigraphy profile is established from **KGI-2024-0417 Rev.2** and **CB-DBM-001 Rev.2** as follows:

- **Design Phreatic Water Table Behind Wall: El. 742.0** (shallow phreatic surface behind the sheet-pile wall, Geotechnical Report **KGI-2024-0417 Rev.2**, Section 7). The excavation side is drained to subgrade El. 726.0 by sumps within the sheeted pit.
- **Confining Clay Aquitard:** Stiff cohesive clay stratum extending from **El. 738.0** to **El. 720.0** (18.0 ft total thickness, leaving a 6.0 ft clay plug below final excavation subgrade El. 726.0).
- **Confined Aquifer Top & Initial Piezometric Head:** The top of the underlying confined dense sand and gravel (SW-GW) aquifer is located at **El. 720.0**, with an initial unpumped confined piezometric head at **El. 738.0** (18.0 ft of artesian head above the aquifer top).

To depressurize the confined SW-GW aquifer prior to deep excavation, four 8-in diameter pressure-relief wells (designated **RW-B1**, **RW-B2**, **RW-B3**, and **RW-B4**) shall be installed outside the sheet line at the four corners of the Block B pit. Each relief well shall be screened from **El. 718.0**

to 705.0 within the dense sand and gravel aquifer and constructed with a continuous annular bentonite-cement grout seal from **El. 721.0 to 738.0** across the clay aquitard interval. The four relief wells shall be actively pumped to hold the confined piezometric head at or below **El. 726.5** throughout excavation and substructure construction, and pumping is maintained until the plant-room slab is cast. The planned drawdown of the confined head is 738.0 − 726.5 = **11.5 ft**.

Three vibrating-wire piezometers (designated **VW-B1**, **VW-B2**, and **VW-B3**, with sensor tips installed at El. 719.0 inside the upper confined aquifer) shall be installed inside the Block B excavation footprint to verify depressurization. The Contractor shall enforce a mandatory **72-hour confirmation hold before excavating below El. 732.0**, during which continuous piezometer readings from VW-B1 through VW-B3 must confirm that the confined aquifer head is maintained at or below **El. 726.5**.

R2.2 — Excavation Base Heave Verification *(Supersedes Rev.1 Section 5)*

With final excavation subgrade at El. 726.0 and the top of the confined dense sand and gravel aquifer at **El. 720.0**, a **6.0 ft clay plug** remains below the excavation base. Base heave stability against artesian uplift is verified in accordance with **DBM §6.2** at the controlled piezometric head of **El. 726.5** (representing a piezometric head h_w = 726.5 – 720.0 = 6.5 ft above the aquifer top at El. 720.0):

- **Unfactored Clay Plug Weight (*W*):** For the **6.0 ft clay plug** at saturated unit weight γ_{clay} = 120 pcf:
  W = 6.0 ft × 120 pcf ⇒ ***W* = 720 psf**
- **Factored Resisting Plug Weight (0.9*W*):** Per **DBM §6.2**, applying the 0.90 resistance factor to total soil plug weight:
  0.9*W* = 648 psf
- **Controlled Upward Water Pressure (*u*):** At the controlled confined head of **El. 726.5** (γ_w = 62.4 pcf):
  u = (726.5 – 720.0) ft × 62.4 pcf = 6.5 ft × 62.4 pcf ⇒ ***u* = 405.6 psf**
- **Base Heave Factor of Safety:**
  FS_{heave} = 0.9W / u = 648 psf / 405.6 psf = 1.60 (**FS 1.60** ≥ 1.50 required per **DBM §6.2** — OK)

R2.3 — Layered Effective-Stress Lateral Loading Above Subgrade *(Supersedes Rev.1 Section 4.3)*

Active lateral earth pressures and hydrostatic water pressures acting on the exposed wall height (H = 24.0 ft, from ground surface El. 750.0 to final subgrade El. 726.0) are computed using a layered effective-stress analysis behind the wall combined with full hydrostatic water pressure below the design water table at **El. 742.0**. Soil unit weights carried from **Submittal 31-SH-B Rev.1** are **118 / 120 / 130 pcf** (118 pcf for the upper granular fill from El. 750.0 to El. 738.0; 120 pcf for the stiff clay aquitard from El. 738.0 to El. 720.0; and 130 pcf for the dense sand and gravel stratum below El. 720.0), with a uniform surface surcharge of **250 psf** per Rev.1 Section 4.1.

Table 2 — Layered Active Earth & Hydrostatic Loading Summary (El. 750.0 to El. 726.0)

Load Component (El. 750.0 to El. 726.0)	Horizontal Line Load (lb/ft)	Moment Arm About Anchor at El. 745.5 (ft)	Driving Moment About El. 745.5 (ft-lb/ft)
Layered Effective-Stress Soil + **250 psf** Surcharge (P_{soil})	11,622.8 lb/ft	10.681 ft (effective resultant)	124,141 ft-lb/ft
Hydrostatic Water Pressure, **El. 742.0** to El. 726.0 (P_w = ½$\gamma_w h_w^2$)	7,987.2 lb/ft	14.167 ft (at El. 731.33)	113,152 ft-lb/ft
Total Horizontal Driving Load & Moment Above Subgrade	**19,610 lb/ft**	—	**237,293 ft-lb/ft**

Total vertical overburden stress behind the wall at subgrade El. 726.0 is:

σ_{v0} = 250 psf + (12.0 ft × 118 pcf) + (12.0 ft × 120 pcf) = 250 + 1,416 + 1,440 = **3,106 psf**

R2.4 — Sheet-Pile Toe Embedment Check *(Supersedes Rev.1 Section 6)*

The steel sheet-pile wall is embedded a depth of **D = 6.5 ft** below final excavation subgrade (El. 726.0) to a sheet-pile tip elevation of **El. 719.5** (total sheet-pile length = 30.5 ft from El. 750.0 to **El. 719.5**; the toe is lowered from the Rev.1 tip elevation of El. 722.0, and Rev.1 Section 13.1, item 3 is amended to the revised tip elevation). Below subgrade the wall is embedded in the CL clay aquitard, whose permeability is of the order of 10^{-8} cm/s, and the embedment is **checked in the undrained condition** using the total-stress net passive pressure formulation (U.S. Steel Sheet Piling Design Manual, anchored bulkhead in clay). The retained side above subgrade is carried drained, with full water pressure, as the conservative envelope. The revised 30.5 ft sheets are driven with the equipment of Rev.1 Section 11.3 (ICE 44B vibratory hammer with Delmag D19 finish), which is adequate for the revised tip elevation.

Using the undrained shear strength s_u = **1,400 psf** carried from **Rev.1 Section 3.2** and the total overburden pressure σ_{v0} = **3,106 psf** at subgrade El. 726.0 from Section **R2.3**, the uniform net passive resistance below subgrade is:

$p_n = 4s_u - \sigma_{v0}$ = 4(1,400 psf) – 3,106 psf = 5,600 psf – 3,106 psf = 2,494.0 psf

Over the embedment depth **D = 6.5 ft**, the unfactored passive toe force and stabilizing moment about the ground anchor elevation at **El. 745.5** are:

- **Unfactored Passive Toe Resistance (P_p):** P_p = 2,494.0 psf × 6.5 ft = 16,211 lb/ft acting at mid-embedment El. 722.75.
- **Moment Arm About Anchor (El. 745.5):** z_p = 745.5 – 722.75 = 22.75 ft.
- **Resisting Passive Moment (M_R):** M_R = 16,211 lb/ft × 22.75 ft = 368,800 ft-lb/ft.
- **Rotational Toe Factor of Safety:** Against the driving moment of **237,293 ft-lb/ft** from Section **R2.3**:
  $FS_{toe} = M_R / M_D$ = 368,800 ft-lb/ft / 237,293 ft-lb/ft = 1.55 (**FS 1.55** ≥ 1.50 required — OK)

The free-earth support formulation adopted presumes full mobilisation of the net passive resistance over the embedded length, consistent with the anchored-bulkhead analysis of the U.S. Steel manual, and the factor of safety of 1.50 applied to the resisting moment covers the mobilisation required.

At tip elevation **El. 719.5**, the sheet-pile toe extends **0.5 ft into the SW-GW** stratum below **El. 720.0** so that the tips bear below the base of the clay, as contemplated in Geotechnical Report Section 9.4. The penetration is nominal and the sheet-pile wall is **not credited as a cutoff** against confined aquifer flow; the aquifer is continuous beneath the pit and is controlled by the relief wells of R2.1, and the sheet interlocks are sealed over the aquitard interval in accordance with Geotechnical Report Section 9.6.

R2.5 — Tieback Ground Anchor Design *(Supersedes Rev.1 Section 7)*

Lateral support near the top of the sheet-pile wall is provided by a single row of multi-strand tieback ground anchors installed at waler elevation **El. 745.5** at an inclination of **20° below horizontal** and spaced horizontally at **6 ft centres**. Each ground anchor is detailed with a **25 ft free length** and a **24 ft bond length** (49 ft total anchor length).

R2.5 — Tieback Ground Anchor Design *(Continued)*

- **Design Anchor Load at 6 ft Centres:** Crediting the allowable passive toe resistance ($P_{p,allow}$ = 16,211 / 1.50 = 10,807 lb/ft) against the total horizontal driving load of **19,610 lb/ft** (Section **R2.3**), the required horizontal anchor reaction is T_h = 19,610 – 10,807 = 8,803 lb/ft. For anchors spaced at **6 ft centres** and inclined at **20° below horizontal**, the design anchor working load is:

  T_{design} = (8,803 lb/ft × 6.0 ft) / cos 20° = 52,818 lb / 0.93969 = 56,208 lb = **56.2 kips**
- **Anchor Bond Capacity:** The **24 ft bond length** is formed by pressure-grouted tieback installation through a 6.0-in nominal diameter drill hole (d_h = 6.0 in) at the design bond stress of **20 psi** adopted in Rev.1 Section 10.4 (FHWA-IF-99-015, Table 4). Temporary drill casing is advanced and grouted over the aquitard interval to seal against artesian seepage during anchor installation. Ultimate bond capacity is:

  $Q_{ult} = \pi \times d_h \times L_b \times \tau_{bond}$ = π × 6.0 in × (24 ft × 12 in/ft) × 20 psi = 108,573 lb = **108.6 kips**
- **Anchor Capacity Check:** Comparing ultimate bond capacity of **108.6 kips** against the design load of **56.2 kips** at **6 ft centres** yields a factor of safety of FS_{anchor} = 108.6 / 56.2 = 1.93 (adequate on the adopted design bond stress of **FHWA-IF-99-015**, Table 4). The anchor tendon is unchanged from Rev.1 Section 10.1: (4) 0.6-in Grade 270 strand, design strength 140 kips against the 56.2 kip design load. Each anchor is proof tested to 1.33 × 56.2 = **74.8 kips** and locked off at 0.75 × 56.2 = **42.2 kips**. The proof load exceeds the design load by 33 per cent and is taken as the demonstration of anchor performance for the period the wall is in service.

R2.6 — Sheet-Pile Structural Bending Check *(Supersedes Rev.1 Section 9)*

The **PZ-35 Grade 50** section adopted in Rev.1 Section 11.2 is retained and re-checked against the revised effective-stress plus hydrostatic lateral loading (F_y = 50 ksi, elastic section modulus S_x = 48.5 in³/ft of wall).

- **Allowable Flexural Capacity (M_{allow}):** At an allowable bending stress of F_b = 0.60 F_y = 30.0 ksi per **AISC 360-16 Chapter F**, as adopted in Rev.1 Section 11.2:

  $M_{allow} = (F_b \times S_x)$ / (12 in/ft) = (30.0 ksi × 48.5 in³/ft) / 12 = **121 kip-ft/ft**
- **Maximum Span Bending Moment (M_{max}):** Under the Section **R2.3** loading and **El. 745.5** anchor reaction, the point of zero shear occurs at **El. 733.1**, producing a maximum flexural demand of **62.4 kip-ft/ft**.
- **Flexural Factor of Safety:** FS_{flex} = 121 kip-ft/ft / 62.4 kip-ft/ft = 1.94 ≥ 1.50 required (**Rev.1 §11.2**) (**PZ-35 Grade 50** section is structurally adequate).

R2.7 — Wall Deflection & Piezometer Trigger *(Supersedes Rev.1 Section 10)*

Wall deflection is re-estimated by the method of Rev.1 Section 12.1 for the **PZ-35 Grade 50** sheet-pile wall tied back at **El. 745.5** under the revised Section **R2.3** loading, giving a predicted maximum lateral wall deflection of **1.3 in** at El. 738.0.

- **Allowable Deflection Criterion:** For exposed excavation height H = 24.0 ft = 288 in, the criterion adopted in **Rev.1 Section 12.3** is **H/200 = 1.44 in** (δ_{max} = **1.3 in** < **H/200 = 1.44 in** — OK).
- **Instrumentation & Construction Action Limit:** Active construction monitoring of piezometers VW-B1 through VW-B3 shall enforce a mandatory **stop-work trigger if the piezometers read above El. 727.0** during excavation below El. 732.0 (in addition to the Section **R2.1** 72-hour confirmation hold). If any piezometer reading exceeds El. 727.0, excavation operations shall cease immediately until relief-well pumping restores piezometric head to or below **El. 726.5**.

R2.8 — Summary of Revised Design Checks & Statement of Adequacy

Table 3 — Summary of Submittal 31-SH-B Revision 2 Supplement Design Verifications

Supplement Section	Design Element / Verification Check	Governing Demand / Condition	Capacity / Allowable Criterion	Status
R2.1	Active Relief-Well Depressurization (4 × 8-in Wells)	Initial Head **El. 738.0**; Water Table **El. 742.0**	Hold ≤ **El. 726.5**; 72-hr hold above El. 732.0	**OK**
R2.2	Excavation Base Heave (**6.0 ft clay plug**, **DBM §6.2**)	Uplift **u = 405.6 psf** at **El. 726.5**	**W = 720 psf**; **0.9W = 648 psf** (**FS 1.60**)	**OK**
R2.3	Layered Effective-Stress + Hydrostatic Loading	**19,610 lb/ft**; M_D = **237,293 ft-lb/ft**	Applied to **R2.4**, **R2.5**, and **R2.6**	**OK**
R2.4	Sheet-Pile Toe Embedment (**D = 6.5 ft** to **El. 719.5**)	M_D = **237,293 ft-lb/ft** (σ_{v0} = **3,106 psf**)	Undrained s_u = **1,400 psf**; M_R = 368,800 (**FS 1.55**)	**OK**
R2.5	Tieback Anchors (**El. 745.5**, **20° below horizontal**)	**56.2 kips** at **6 ft centres** (25 ft / 24 ft)	**108.6 kips** at **20 psi** pressure-grouted bond (FS = 1.93)	**OK**
R2.6	Sheet-Pile Flexure (**PZ-35 Grade 50**)	M_{max} = **62.4 kip-ft/ft** at **El. 733.1**	M_{allow} = **121 kip-ft/ft** (FS = 1.94)	**OK**
R2.7	Lateral Wall Deflection & Piezometer Trigger	Predicted Deflection = **1.3 in**	Allowable **H/200 = 1.44 in**; Trigger > El. 727.0	**OK**

Conclusion of Adequacy: Based on the revised engineering analyses presented in Sections **R2.1** through **R2.7** above, Kesterline Engineering, PC concludes that the Block B plant-room support of excavation design in **Submittal 31-SH-B Rev.1**, as modified by this Revision 2 supplement (**KE-L-2026-0713**), conforms to **KGI-2024-0417 Rev.2**, **CB-DBM-001 Rev.2**, **Section 31 20 00 Rev.3**, and the **Package A register**, and is geotechnically and structurally adequate for construction. Relief-well installation, the 72-hour piezometer confirmation and the anchor proof tests are hold points in the Contractor's sequence under Rev.1 Section 13.1. We request verification of Submittal 31-SH-B on the basis of Rev.1 as supplemented by Rev.2, so that excavation can proceed on the Contractor's planned date.

CERTIFICATION & PROFESSIONAL ENGINEER SEAL

Respectfully submitted,

Kesterline Engineering, PC

/s/ Byron A. Halvard

Byron A. Halvard, PE — Principal Engineer, Kesterline Engineering, PC
Ohio Professional Engineer License No. **E-58730** — sealed 07/13/2026
Prepared by: R. T. Karcher, EIT Checked by: S. M. Ovrud, PE

[PROFESSIONAL ENGINEER SEAL — B. A. HALVARD, PE — STATE OF OHIO LICENSE NO. E-58730 — 07/13/2026]